

```
In [1]: import os
        print(os.listdir())
```

```
Out:
['.ipynb_checkpoints', 'test.csv', 'Untitled.ipynb']
```

```
In [2]: import pandas as pd

        df = pd.read_csv("test.csv")
        print(df.head())
        print(df.columns)
```

```
Out:
          tweets      class
0  no one ever predicted this was going to happen...  figurative
1  @Stooshie its as closely related as Andrews or...  figurative
2  I find it ironic when Vegans say they love foo...  figurative
3  Quick rt that throwing money vine I've not see...  figurative
4  yep, keep adding me to your #devops lists.... ...  figurative
Index(['tweets', 'class'], dtype='object')
```

```
In [3]: df.info()
```

```
Out:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8128 entries, 0 to 8127
Data columns (total 2 columns):
 #   Column  Non-Null Count  Dtype  
---  -
 0   tweets  8126 non-null    object 
 1   class   8119 non-null    object 
dtypes: object(2)
memory usage: 127.1+ KB
```

```
In [4]: import pandas as pd
```

```
df = pd.read_csv("test.csv")

print("Shape:", df.shape)
print("\nColumns:")
print(df.columns.tolist())

print("\nFirst 5 Rows:")
print(df.head())
```

Out:

Shape: (8128, 2)

Columns:

['tweets', 'class']

First 5 Rows:

	tweets	class
0	no one ever predicted this was going to happen...	figurative
1	@Stooshie its as closely related as Andrews or...	figurative
2	I find it ironic when Vegans say they love foo...	figurative
3	Quick rt that throwing money vine I've not see...	figurative
4	yep, keep adding me to your #devops lists....	figurative

In [5]: pip install vaderSentiment

Out:

Collecting vaderSentiment

Downloading vaderSentiment-3.3.2-py2.py3-none-any.whl.metadata (572 bytes)

Requirement already satisfied: requests in c:\users\user\anaconda2\lib\site-packages (from vaderSentiment) (2.32.5)

Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\user\anaconda2\lib\site-packages (from requests->vaderSentiment) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in c:\users\user\anaconda2\lib\site-packages (from requests->vaderSentiment) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\user\anaconda2\lib\site-packages (from requests->vaderSentiment) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in c:\users\user\anaconda2\lib\site-packages (from requests->vaderSentiment) (2025.11.12)

Downloading vaderSentiment-3.3.2-py2.py3-none-any.whl (125 kB)

Installing collected packages: vaderSentiment

Successfully installed vaderSentiment-3.3.2

Note: you may need to restart the kernel to use updated packages.

```
In [6]: import pandas as pd
        from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer

df = pd.read_csv("test.csv")

analyzer = SentimentIntensityAnalyzer()

# Replace 'text' with your actual tweet column name
text_column = 'text'

def get_sentiment(text):
    score = analyzer.polarity_scores(str(text))['compound']

    if score >= 0.05:
        return "Positive"
    elif score <= -0.05:
        return "Negative"
    else:
        return "Neutral"

df['Sentiment'] = df[text_column].apply(get_sentiment)

print(df[[text_column, 'Sentiment']].head())

print("\nSentiment Counts:")
print(df['Sentiment'].value_counts())
```

Out:

```
-----
KeyError                                Traceback (most recent call last)
File ~\anaconda2\Lib\site-packages\pandas\core\indexes\base.py:3812, in Index.get_loc(self, key)
    3811 try:
-> 3812     return self._engine.get_loc(casted_key)
    3813 except KeyError as err:

File pandas\_libs\index.pyx:167, in pandas._libs.index.IndexEngine.get_loc()

File pandas\_libs\index.pyx:196, in pandas._libs.index.IndexEngine.get_loc()
```

```
File pandas/_libs/hashtable_class_helper.pxi:7088, in pandas._libs.hashtable.PyObjectHashTable.get_item()
e.get_item()
```

```
File pandas/_libs/hashtable_class_helper.pxi:7096, in pandas._libs.hashtable.PyObjectHashTable.get_item()
e.get_item()
```

```
KeyError: 'text'
```

The above exception was the direct cause of the following exception:

```
KeyError                                Traceback (most recent call last)
```

```
Cell In[6], line 21
```

```
    18     else:
    19         return "Neutral"
--> 21 df['Sentiment'] = df[text_column].apply(get_sentiment)
    23 print(df[[text_column, 'Sentiment']].head())
    25 print("\nSentiment Counts:")
```

```
File ~\anaconda2\Lib\site-packages\pandas\core\frame.py:4113, in DataFrame.__getitem__(self, key)
4111 if self.columns.nlevels > 1:
4112     return self._getitem_multilevel(key)
-> 4113 indexer = self.columns.get_loc(key)
4114 if is_integer(indexer):
4115     indexer = [indexer]
```

```
4111 if self.columns.nlevels > 1:
4112     return self._getitem_multilevel(key)
-> 4113 indexer = self.columns.get_loc(key)
4114 if is_integer(indexer):
4115     indexer = [indexer]
```

```
File ~\anaconda2\Lib\site-packages\pandas\core\indexes\base.py:3819, in Index.get_loc(self, key)
3817 if isinstance(casted_key, slice) or (
3818     isinstance(casted_key, abc.Iterable)
3819     and any(isinstance(x, slice) for x in casted_key)
3820 ):
3821     raise InvalidIndexError(key)
-> 3819 raise KeyError(key) from err
3820 except TypeError:
```

```
3814     if isinstance(casted_key, slice) or (
3815         isinstance(casted_key, abc.Iterable)
3816         and any(isinstance(x, slice) for x in casted_key)
3817     ):
3818         raise InvalidIndexError(key)
-> 3819 raise KeyError(key) from err
3820 except TypeError:
3821     # If we have a listlike key, _check_indexing_error will raise
3822     # InvalidIndexError. Otherwise we fall through and re-raise
3823     # the TypeError.
3824     self._check_indexing_error(key)
```

```
KeyError: 'text'
```

```
In [7]: import matplotlib.pyplot as plt
```

```

sentiment_counts = df['Sentiment'].value_counts()

plt.figure(figsize=(6,4))
sentiment_counts.plot(kind='bar')
plt.title('Sentiment Analysis of Tweets')
plt.xlabel('Sentiment')
plt.ylabel('Number of Tweets')
plt.show()

```

Out:

```

-----
KeyError                                Traceback (most recent call last)
File ~\anaconda2\Lib\site-packages\pandas\core\indexes\base.py:3812, in Index.get_loc(self, key)
    3811 try:
-> 3812     return self._engine.get_loc(casted_key)
    3813 except KeyError as err:

File pandas\_libs\index.pyx:167, in pandas._libs.index.IndexEngine.get_loc()

File pandas\_libs\index.pyx:196, in pandas._libs.index.IndexEngine.get_loc()

File pandas\_libs\hashtable_class_helper.pxi:7088, in pandas._libs.hashtable.PyObjectHashTable.get_item()

File pandas\_libs\hashtable_class_helper.pxi:7096, in pandas._libs.hashtable.PyObjectHashTable.get_item()

KeyError: 'Sentiment'

```

The above exception was the direct cause of the following exception:

```

KeyError                                Traceback (most recent call last)
Cell In[7], line 3
      1 import matplotlib.pyplot as plt
----> 3 sentiment_counts = df['Sentiment'].value_counts()
      5 plt.figure(figsize=(6,4))
      6 sentiment_counts.plot(kind='bar')

File ~\anaconda2\Lib\site-packages\pandas\core\frame.py:4113, in DataFrame.__getitem__(self, key)
    4111 if self.columns.nlevels > 1:
    4112     return self._getitem_multilevel(key)
-> 4113 indexer = self.columns.get_loc(key)
    4114 if is_integer(indexer):

```

```

4115     indexer = [indexer]

File ~\anaconda2\Lib\site-packages\pandas\core\indexes\base.py:3819, in Index.get_loc(self, key)
3814     if isinstance(casted_key, slice) or (
3815         isinstance(casted_key, abc.Iterable)
3816         and any(isinstance(x, slice) for x in casted_key)
3817     ):
3818         raise InvalidIndexError(key)
-> 3819     raise KeyError(key) from err
3820 except TypeError:
3821     # If we have a listlike key, _check_indexing_error will raise
3822     # InvalidIndexError. Otherwise we fall through and re-raise
3823     # the TypeError.
3824     self._check_indexing_error(key)

KeyError: 'Sentiment'

```

```

In [8]: import pandas as pd

df = pd.read_csv("test.csv")

print(df.columns.tolist())

```

```

Out:
['tweets', 'class']

```

```

In [9]: print(df.head())

```

```

Out:

```

	tweets	class
0	no one ever predicted this was going to happen...	figurative
1	@Stooshie its as closely related as Andrews or...	figurative
2	I find it ironic when Vegans say they love foo...	figurative
3	Quick rt that throwing money vine I've not see...	figurative
4	yep, keep adding me to your #devops lists....	figurative

```

In [10]: text_column = 'Tweet'

```

```
In [11]: text_column = 'tweet'
```

```
In [12]: text_column = 'text'
```

```
In [13]: print(df.columns.tolist())
```

Out:

```
['tweets', 'class']
```

```
In [14]: import pandas as pd
```

```
df = pd.read_csv("test.csv")
```

```
print(df.columns)
```

Out:

```
Index(['tweets', 'class'], dtype='object')
```

```
In [15]: print(df.head())
```

Out:

	tweets	class
0	no one ever predicted this was going to happen...	figurative
1	@Stooshie its as closely related as Andrews or...	figurative
2	I find it ironic when Vegans say they love foo...	figurative
3	Quick rt that throwing money vine I've not see...	figurative
4	yep, keep adding me to your #devops lists.... ...	figurative

```
In [16]: text_column = 'text'
```

```
In [17]: text_column = 'message'
```

```
In [18]: print(df.columns.tolist())
```

Out:

```
['tweets', 'class']
```

Install VADER (run once if needed)

pip install vaderSentiment

```
import pandas as pd from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer import matplotlib.pyplot as plt
```

Load dataset

```
df = pd.read_csv("test.csv")
```

Display columns

```
print("Columns in dataset:") print(df.columns.tolist())
```

Replace this with the actual tweet/text column name from the output above

```
text_column = input("Enter the tweet/text column name exactly as shown above: ")
```

Initialize VADER

```
analyzer = SentimentIntensityAnalyzer()
```


Function to classify sentiment

```
def get_sentiment(text): score = analyzer.polarity_scores(str(text))['compound']
```

```
    if score >= 0.05:  
        return "Positive"  
    elif score <= -0.05:  
        return "Negative"  
    else:  
        return "Neutral"
```

Apply sentiment analysis

```
df['Sentiment'] = df[text_column].apply(get_sentiment)
```

Display results

```
print("\nFirst 5 Results:") print(df[[text_column, 'Sentiment']].head())
```

Sentiment counts

```
print("\nSentiment Counts:") print(df['Sentiment'].value_counts())
```

Bar Chart

```
plt.figure(figsize=(6,4)) df['Sentiment'].value_counts().plot(kind='bar') plt.title('Sentiment Analysis of  
Tweets') plt.xlabel('Sentiment') plt.ylabel('Count') plt.show()
```

Pie Chart

```
plt.figure(figsize=(6,6)) df['Sentiment'].value_counts().plot( kind='pie', autopct='%1.1f%%' )  
plt.title('Tweet Sentiment Distribution') plt.ylabel('') plt.show()
```

Save results

```
df.to_csv("tweets_sentiment_results.csv", index=False)
```

```
print("\nSentiment analysis completed successfully!") print("Results saved as: tweets_sentiment_results.csv")
```

```
In [19]: print(df.columns.tolist())
```

Out:

```
['tweets', 'class']
```

```
In [20]: import pandas as pd
```

```
df = pd.read_csv("test.csv")
```

```
print(df.columns.tolist())
```

```
print(df.head())
```

Out:

```
['tweets', 'class']
```

	tweets	class
0	no one ever predicted this was going to happen...	figurative
1	@Stooshie its as closely related as Andrews or...	figurative
2	I find it ironic when Vegans say they love foo...	figurative
3	Quick rt that throwing money vine I've not see...	figurative
4	yep, keep adding me to your #devops lists....	figurative

```
In [21]: print(df.columns.tolist())
```

```
print(df.head())
```

Out:

```
['tweets', 'class']
```

	tweets	class
0	no one ever predicted this was going to happen...	figurative
1	@Stooshie its as closely related as Andrews or...	figurative

```

2 I find it ironic when Vegans say they love foo... figurative
3 Quick rt that throwing money vine I've not see... figurative
4 yep, keep adding me to your #devops lists.... ... figurative

```

In [22]: `import matplotlib.pyplot as plt`

```

sentiment_counts = df['Sentiment'].value_counts()

plt.figure(figsize=(8,5))
sentiment_counts.plot(kind='bar')
plt.title('Sentiment Distribution')
plt.xlabel('Sentiment')
plt.ylabel('Number of Tweets')
plt.xticks(rotation=0)
plt.show()

```

Out:

```

-----
KeyError                                Traceback (most recent call last)
File ~\anaconda2\Lib\site-packages\pandas\core\indexes\base.py:3812, in Index.get_loc(self, k
ey)
    3811 try:
-> 3812     return self._engine.get_loc(casted_key)
    3813 except KeyError as err:

File pandas/_libs/index.pyx:167, in pandas._libs.index.IndexEngine.get_loc()

File pandas/_libs/index.pyx:196, in pandas._libs.index.IndexEngine.get_loc()

File pandas/_libs/hashtable_class_helper.pxi:7088, in pandas._libs.hashtable.PyObjectHashTabl
e.get_item()

File pandas/_libs/hashtable_class_helper.pxi:7096, in pandas._libs.hashtable.PyObjectHashTabl
e.get_item()

KeyError: 'Sentiment'

The above exception was the direct cause of the following exception:

KeyError                                Traceback (most recent call last)
Cell In[23], line 3
      1 import matplotlib.pyplot as plt
----> 3 sentiment_counts = df['Sentiment'].value_counts()

```

```
5 plt.figure(figsize=(8,5))
6 sentiment_counts.plot(kind='bar')
```

File ~\anaconda2\Lib\site-packages\pandas\core\frame.py:4113, in DataFrame.__getitem__(self, key)

```
4111 if self.columns.nlevels > 1:
4112     return self._getitem_multilevel(key)
-> 4113 indexer = self.columns.get_loc(key)
4114 if is_integer(indexer):
4115     indexer = [indexer]
```

File ~\anaconda2\Lib\site-packages\pandas\core\indexes\base.py:3819, in Index.get_loc(self, key)

```
3814     if isinstance(casted_key, slice) or (
3815         isinstance(casted_key, abc.Iterable)
3816         and any(isinstance(x, slice) for x in casted_key)
3817     ):
3818         raise InvalidIndexError(key)
-> 3819     raise KeyError(key) from err
3820 except TypeError:
3821     # If we have a listlike key, _check_indexing_error will raise
3822     # InvalidIndexError. Otherwise we fall through and re-raise
3823     # the TypeError.
3824     self._check_indexing_error(key)
```

KeyError: 'Sentiment'

In [23]: `print(df.head())`

Out:

	tweets	class
0	no one ever predicted this was going to happen...	figurative
1	@Stooshie its as closely related as Andrews or...	figurative
2	I find it ironic when Vegans say they love foo...	figurative
3	Quick rt that throwing money vine I've not see...	figurative
4	yep, keep adding me to your #devops lists....	figurative

```
In [24]: import pandas as pd
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer
import matplotlib.pyplot as plt
import seaborn as sns
```

```

# Load dataset
df = pd.read_csv("test.csv")

# Initialize VADER
analyzer = SentimentIntensityAnalyzer()

# Sentiment function
def get_sentiment(tweet):
    score = analyzer.polarity_scores(str(tweet))['compound']

    if score >= 0.05:
        return 'Positive'
    elif score <= -0.05:
        return 'Negative'
    else:
        return 'Neutral'

# Create Sentiment column
df['Sentiment'] = df['tweets'].apply(get_sentiment)

# Show first 5 results
print(df[['tweets', 'Sentiment']].head())

# Count sentiments
print("\nSentiment Counts:")
print(df['Sentiment'].value_counts())

```

Out:

	tweets	Sentiment
0	no one ever predicted this was going to happen...	Negative
1	@Stooshie its as closely related as Andrews or...	Positive
2	I find it ironic when Vegans say they love foo...	Positive
3	Quick rt that throwing money vine I've not see...	Negative
4	yep, keep adding me to your #devops lists....	Positive

Sentiment Counts:

Sentiment

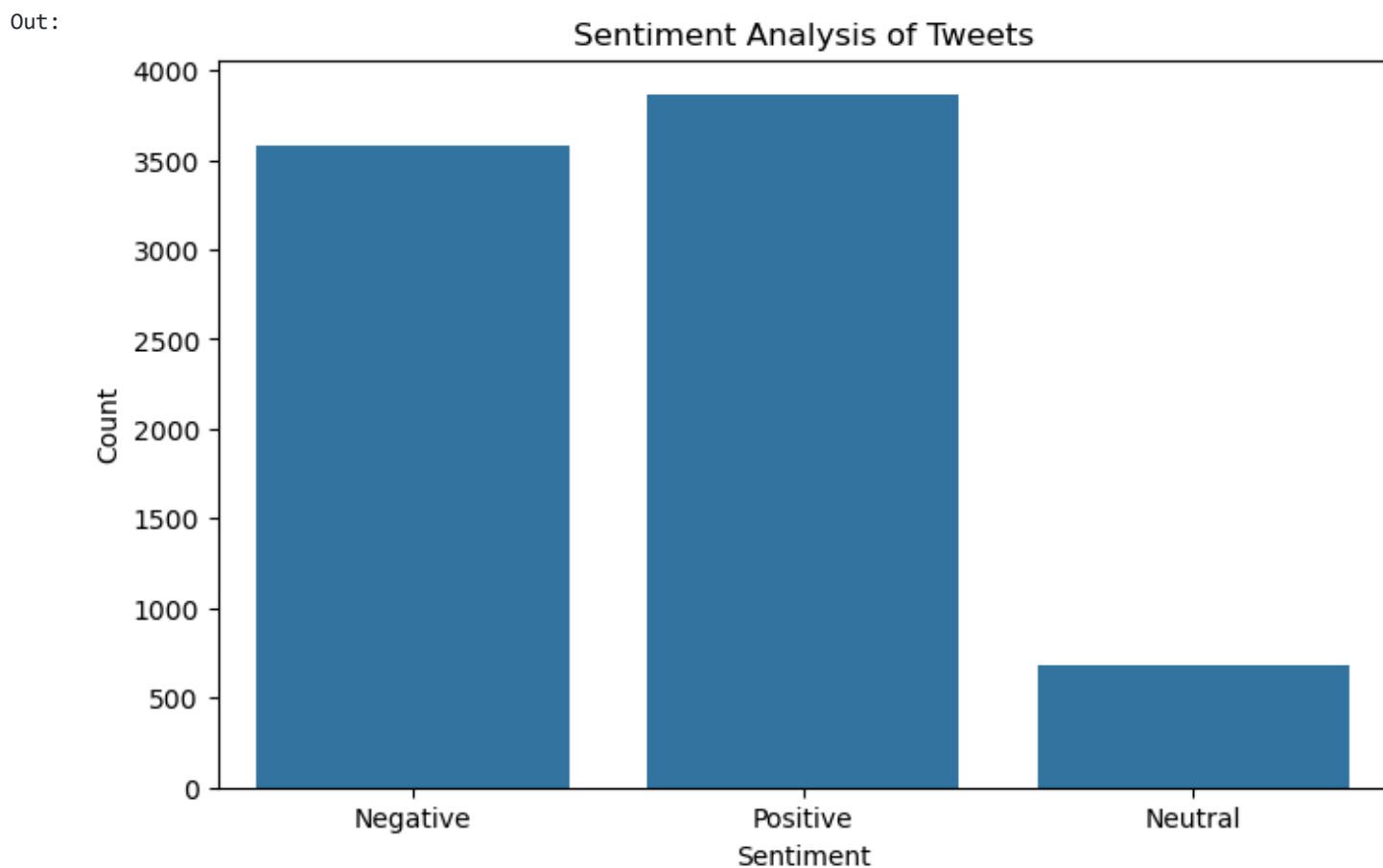
Positive 3864

Negative 3581

Neutral 683

Name: count, dtype: int64

```
In [25]: plt.figure(figsize=(8,5))
sns.countplot(data=df, x='Sentiment')
plt.title('Sentiment Analysis of Tweets')
plt.xlabel('Sentiment')
plt.ylabel('Count')
plt.show()
```



```
In [26]: df.to_csv("tweets_sentiment_results.csv", index=False)
print("Results saved successfully!")
```

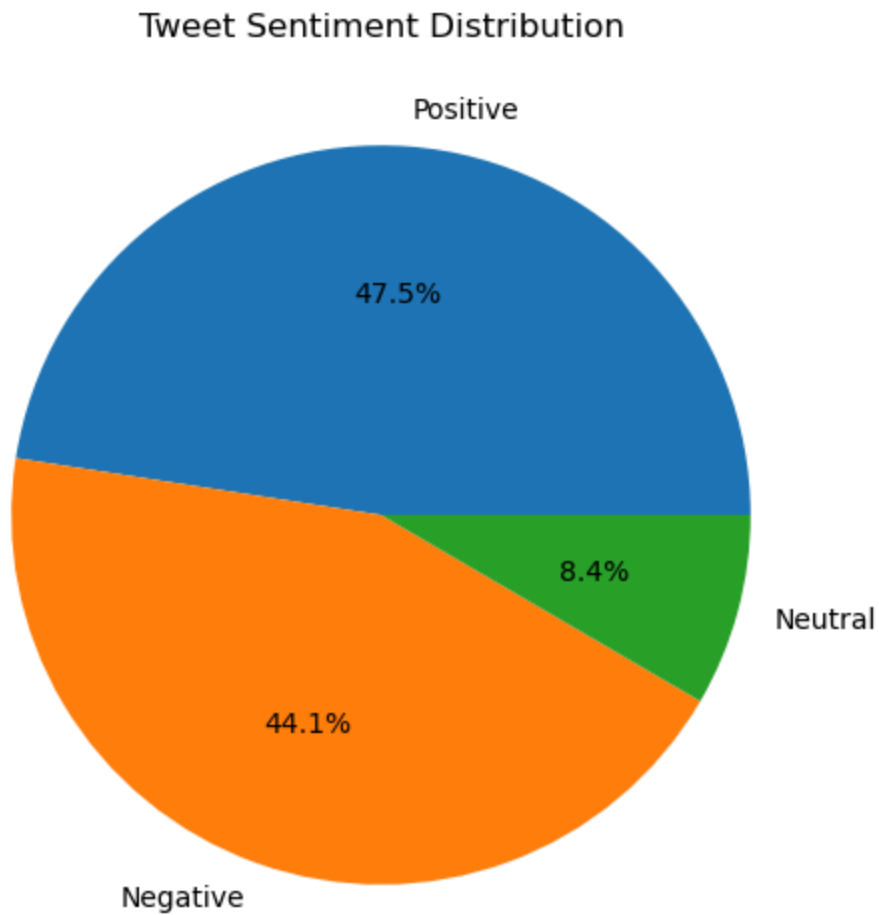
Out:

```
Results saved successfully!
```

```
In [27]: plt.figure(figsize=(6,6))
df['Sentiment'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%'
```

```
)  
plt.title('Tweet Sentiment Distribution')  
plt.ylabel('')  
plt.show()
```

Out:



```
In [28]: sentiment_percentage = df['Sentiment'].value_counts(normalize=True) * 100
```

```
print("Sentiment Percentage:")  
print(sentiment_percentage)
```

Out:

```
Sentiment Percentage:  
Sentiment  
Positive    47.539370  
Negative    44.057579  
Neutral      8.403051  
Name: proportion, dtype: float64
```

```
In [29]: summary = pd.DataFrame({
    'Count': df['Sentiment'].value_counts(),
    'Percentage': round(df['Sentiment'].value_counts(normalize=True) * 100, 2)
})

print(summary)
```

Out:

	Count	Percentage
Sentiment		
Positive	3864	47.54
Negative	3581	44.06
Neutral	683	8.40

```
In [30]: negative_tweets = df[df['Sentiment'] == 'Negative']

print("Top Negative Tweets:")
print(negative_tweets['tweets'].head(10))
```

Out:

```
Top Negative Tweets:
0    no one ever predicted this was going to happen...
3    Quick rt that throwing money vine I've not see...
6           @RalphtheBold: But ... football. #sarcasm
7    So last month I lost my job and today my dad l...
8    Rahm has solved crime, budget & pension is...
10   This girl can't wait to shop on Michigan Ave. ...
12   @Nola2016 ...says the person typing in shortha...
13   Crowded hallways are the loneliest places #irony
15   I'll go for @NazranaYusufzai to say that. #sar...
16           oh wait are you even in this episode? #ironic
Name: tweets, dtype: object
```

```
In [31]: positive_tweets = df[df['Sentiment'] == 'Positive']

print("Top Positive Tweets:")
print(positive_tweets['tweets'].head(10))
```


Out:

Top Positive Tweets:

```
1    @Stooshie its as closely related as Andrews or...
2    I find it ironic when Vegans say they love foo...
4    yep, keep adding me to your #devops lists.... ...
5    wonderful time with customer service @TMobile ...
9    And the heel on my @katespadeny shoe just brok...
11   That shitty feeling we all love so much 😊 #sar...
14   @dorkiesaurus I remember when I lived in Downt...
18   Maybe it's just me, but I'm sure that's not si...
21   I just love getting anxiety at random times. I...
23   To Liberals who care so much about Corporation...
```

Name: tweets, dtype: object

In [32]: `pip install wordcloud`